

Existence, Uniqueness, and Construction of Haar Measure on Locally Compact Hausdorff Topological Groups

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Background Information

Definition:

A **Topological Group** is a group $\langle G, \cdot \rangle$ with a topology τ such that the following maps are continuous:

- ▶ $\varphi : G \rightarrow G$ defined by $\varphi : x \mapsto x^{-1}$.
- ▶ $\psi : G \times G \rightarrow G$ defined by $\psi : (a, b) \mapsto a \cdot b$.

Definition:

Let $f : X \rightarrow \mathbb{R}$. The **support** of f , denoted $\text{supp}(f)$ is the set $\overline{\{x \in X : f(x) \neq 0\}}$. That is, it is the set of domain points that are not mapped to 0.

Definition:

Let G be a topological group. The set of **continuous functions with compact support on G** is denoted by:

$$C_c(G) = \{f : G \rightarrow \mathbb{R} : f \text{ is continuous and } \text{supp}(f) \text{ is compact}\}.$$

Haar and Radon Measures

Definition:

A measure m is **inner regular** if for any open set U , $m(U) = \sup\{m(K) : K \text{ is a compact subset of } U\}$.

Definition:

A measure m is **outer regular** if for any measurable set E , $m(E) = \inf\{m(U) : E \subseteq U, U \text{ is open}\}$.

Definition:

A measure m is **locally finite** if for any x , there exists an open set U containing x such that $m(U) < \infty$.

Definition:

A **Radon Measure** is a measure m that is inner regular, outer regular and locally finite.

Definition:

A left **Haar Measure** on a locally compact group G is a nonzero, left invariant radon measure on G .

"Quasi" Measure Approximation

Definition:

Let $\mathcal{B}_G$ denote the σ -algebra generated by τ . If $E \in \mathcal{B}_G$ and V is open and nonempty, then:

$$(E : V) = \inf \{ |A| : E \subset \bigcup_{x \in A} xV \}.$$

($|\cdot|$ denotes cardinality). Thus, $(E : V)$ is the smallest number of left translates of V that cover E .

Examples

Example in $\mathbb{R}$

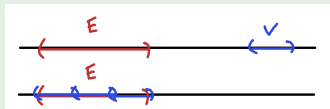


Figure 1: Example in $\mathbb{R}$ with $(E : V) = 3$.

"Quasi" Measure Approximation

Notice that if we predefine the size of a bounded open set F to be 1, then the following is an approximation of the size of E :

$$S(E) \approx \frac{(E : V)}{(F : V)}.$$

Notice that as V gets smaller, $S(E)$ becomes more accurate. We will show that $S(E)$ becomes a Haar Measure as V shrinks to $\{e\}$.

Definition:

Let $f : X \rightarrow \mathbb{R}$. The **Uniform Norm**, denoted $\|f\|_u$ or $\|f\|_\infty$ is defined as follows:

$$\|f\|_u = \|f\|_\infty = \sup_X \{|f|\}.$$

Definition:

We will use C_c^+ to denote $\{f \in C_c(G) : f \geq 0 \text{ and } \|f\|_\infty > 0\}$.

In Terms of Integrals..

Rather than working with measures of sets, we will consider integrals of functions. Let $f, \phi \in C_c^+$. Then $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is open and nonempty.

Proof.

We must show that $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is open and nonempty. Suppose FSO that $\{\phi(x) : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is empty. Thus, $\phi(x) \leq \frac{1}{2}\|\phi\|_\infty$ for all $x \in G$. This gives that

$$\phi(x) \leq \frac{1}{2}\|\phi\|_\infty < \frac{3}{4}\|\phi\|_\infty < \|\phi\|_\infty.$$

Hence, $\frac{3}{4}\|\phi\|_\infty$ is a lower upper bound than $\sup\{\phi\}$, which is a contradiction. So, $\{\phi(x) : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is nonempty and so $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is nonempty. $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\} = \phi^{-1}\{(\frac{1}{2}\|\phi\|_\infty, \infty)\}$ and therefore is open. $\square$

In Terms of Integrals..

We have shown that $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ is open and nonempty, and so finitely many left translates of it cover $\text{supp}(f)$. (In essence, it is functioning similarly to the V from earlier) As a result,

$$f(x) \leq \frac{2\|f\|_\infty}{\|\phi\|_\infty} \sum_{j \leq n} L_{x_j} \phi(x).$$

Proof.

Suppose $x \in \text{supp}(f)$. We have just shown that finitely many left translates of $\{x : \phi(x) > \frac{1}{2}\|\phi\|_\infty\}$ cover $\text{supp}(f)$, so there exists $j \in \mathbb{N}$ such that $L_{x_j} \phi(x) > \frac{1}{2}\|\phi\|_\infty$. So,

$$\sum_{j \leq n} L_{x_j} \phi(x) > \frac{1}{2}\|\phi\|_\infty \implies \frac{2\|f\|_\infty}{\|\phi\|_\infty} \sum_{j \leq n} L_{x_j} \phi(x) > \|f\|_\infty \geq f(x).$$

If $x \notin \text{supp}(f)$ then the result is obvious. □

The Haar Covering Number

Definition:

We call

$$\inf \left\{ \sum_{j \leq n} c_j : f \leq \sum_{j \leq n} c_j L_{x_j} \phi \right\}$$

the **Haar Covering Number** of f with respect to ϕ and denote it $(f : \phi)$. Note that previously $(* : *)$ was defined in terms of sets, and now we are defining it in terms of functions.

Properties:

1. $(f : \phi) = (L_x f : \phi)$ for any $x \in G$.
2. $(cf : \phi) = c(f : \phi)$ for any $c > 0$.
3. $(f + g : \phi) \leq (f : \phi) + (g : \phi)$.
4. $(f : \phi) \leq (f : g)(g : \phi)$.

Normalization

We now make a normalization by choosing $f_0 \in C_c^+$ to have measure 1 and define the following:

$$I_\phi(f) = \frac{(f : \phi)}{(f_0 : \phi)}, \text{ for } f, \phi \in C_c^+.$$

Note that by properties 1-3, for a fixed ϕ , the functional I_ϕ is left-invariant and subadditive. We will now show that I_ϕ becomes approximately additive whenever $\text{supp}(\phi)$ becomes small.

Lemma:

If $f_1, f_2 \in C_c^+$ and $\epsilon > 0$, there is a neighborhood V of e such that $I_\phi(f_1) + I_\phi(f_2) \leq I_\phi(f_1 + f_2) + \epsilon$ whenever $\text{supp}(\phi) \subseteq V$.

Approximate Additivity of I_ϕ

Proof.

Fix $g \in C_c^+$ such that $g\{\text{supp}(f_1 + f_2)\} = 1$ and let $\delta > 0$. Let $h = f_1 + f_2 + \delta g$, and $h_i = \frac{f_i}{h}$ ($i = 1, 2$). Then $h_i \in C_c^+$, and there is a neighborhood V of e such that $|h_i(x) - h_i(y)| < \delta$ if $y^{-1}x \in V$. If $\phi \in C_c^+$, $\text{supp}(\phi) \subset V$, and $h \leq \sum_{j \leq n} c_j L_{x_j} \phi$, then $|h_i(x) - h_i(x_j)| < \delta$ whenever $x_j^{-1}x \in \text{supp}(\phi)$, so

$$f_i(x) = h(x)h_i(x) \leq \sum_j c_j \phi(x_j^{-1}x)h_i(x) \leq \sum_j c_j \phi(x_j^{-1}x)[h_i(x_j) + \delta].$$

But then $(f_i : \phi) \leq \sum c_j [h_i(x_j) + \delta]$, and since $h_1 + h_2 \leq 1$.

$$(f_1 : \phi) + (f_2 : \phi) \leq \sum_j c_j [1 + 2\delta].$$

Now, $\sum c_j$ can be made arbitrarily close to $(h : \phi)$, so

$$I_\phi(f_1) + I_\phi(f_2) \leq (1 + 2\delta)I_\phi(h) \leq (1 + 2\delta)[I_\phi(f_1 + f_2) + \delta I_\phi(g)].$$

Thus, choosing sufficiently small δ :

$$2\delta(f_1 + f_2 : f_0) + \delta(1 + 2\delta)(g : f_0) < \epsilon.$$



Preliminary Theorems

Tychonoff's Theorem

If $\{X_i\}_{i \in I}$ is a family of compact topological spaces, then the product space $X = \prod_{i \in I} X_i$ is compact under the product topology.

The Riesz-Markov-Kakutani Representation Theorem

Let X be a locally compact Hausdorff space and I a positive linear functional on $C_c(X)$. Then there exists a unique positive Borel measure μ on X such that:

$$I(f) = \int_X f d\mu, \quad \forall f \in C_c(X).$$

Existence of Haar Measure on Locally Compact Groups

Theorem. Every locally compact group G has a left Haar measure.

Proof Outline:

1. We define the space that contains all of the I_ϕ s. This is the product space of the possible functional values for each $f \in C_c^+$.
2. Show that there is an element in the intersection of the closure of all compact neighborhoods of e .
3. Show that the element within the intersection is a positive linear functional on C_c^+ .
4. Extend the positive linear functional to C_c and show left-invariance.
5. Use Riesz–Markov–Kakutani Representation Theorem to show that the functional is a Haar measure.

Proof of Existence

Proof.

1. Define $X_f = [(f_0 : f)^{-1}], (f : f_0)]$ for each $f \in C_c^+$. Let $X = \prod_{f \in C_c^+} X_f$. Then because each X_f is compact, the product X is compact and Hausdorff by Tychonoff's Theorem. Furthermore, each $l_\phi \in X$ by properties of the Haar Covering Number.
2. Let V be a compact neighborhood of e , and $K(V)$ be $\overline{\{l_\phi : \text{supp}(\phi) \in V\}}$. This is the closure of the set of functionals of functions with support contained in V . It follows that $K(\bigcap_{j \leq n} V_j) \subseteq \bigcap_{j \leq n} K(V_j)$. Because $\bigcap_{j \leq n} V_j$ is a neighborhood of the identity, so is $K(\bigcap_{j \leq n} V_j)$. Thus, there is an element $l \in \bigcap_{j \leq n} K(V_j)$.
3. By approximate additivity, for any $f_n \in C_c^+$ and $\epsilon > 0$, there is $\phi \in C_c^+$ with $\text{supp}(\phi) \subseteq V$ such that $|l(f_j) - l_\phi(f_j)| < \epsilon$. Thus, l is left-invariant and $l(af + bg) = al(f) + bl(g)$. Thus, l is a positive linear functional on C_c^+ .

Proof of Existence Cont.

Proof.

4. We can extend I to C_c similarly to how we extended the Lebesgue integral beyond positive functions. Let $I(f) = I(f^+) - I(f^-)$. We have now extended I to C_c , and it remains left-invariant. $I(f) > 0$ for all $f \in C_c^+$, because of our definition for X_f and X .
5. It follows by the Riesz-Markov-Kakutani Representation Theorem that I is a left Haar measure.



Uniqueness (for abelian groups)

Theorem.

Let μ and ν be left Haar measures on G . Then there exists $c > 0$ such that $\mu = c\nu$.

Proof.

Note if G is abelian, then μ is both left and right invariant. Let $h \in C_c^+$ such that $h(x) = h(x^{-1})$. Let $f \in C_c(G)$. Then

$$\begin{aligned}\int_G h d\nu \int_G f d\mu &= \iint_{G \times G} h(y)f(x) d\mu(x) d\nu(y). \\ &= \iint_{G \times G} h(y)f(xy) d\mu(x) d\nu(y). \text{ (by right invariance)} \\ &= \iint_{G \times G} h(y)f(xy) d\nu(y) d\mu(x). \text{ (by Fubini/Tonelli)} \\ &= \iint_{G \times G} h(x^{-1}y)f(y) d\nu(y) d\mu(x) = \iint_{G \times G} h(y^{-1}x)f(y) d\nu(y) d\mu(x). \\ &= \iint_{G \times G} h(y^{-1}x)f(y) d\mu(x) d\nu(y) = \iint_{G \times G} h(x)f(y) d\mu(x) d\nu(y). \\ &= \int_G h d\mu \int_G f d\nu. \text{ so, } \frac{\int h d\nu}{\int h d\mu} = \frac{\int f d\nu}{\int f d\mu}.\end{aligned}$$



Examples of Haar Measures

Examples

1. Let $G = \langle \mathbb{R}^+, \times \rangle$. Then $\frac{dx}{x}$ is a Haar measure on G .
2. Let $G = \langle \mathbb{R}, + \rangle$. Then Lebesgue measure is a Haar measure on G .
3. Let $G = \langle \{z \in \mathbb{C} : |z| = 1\}, \times \rangle$. Then arc length is a Haar measure on G .
4. let $G = \langle \mathbb{Z}, + \rangle$. Then counting measure is a Haar measure on G .